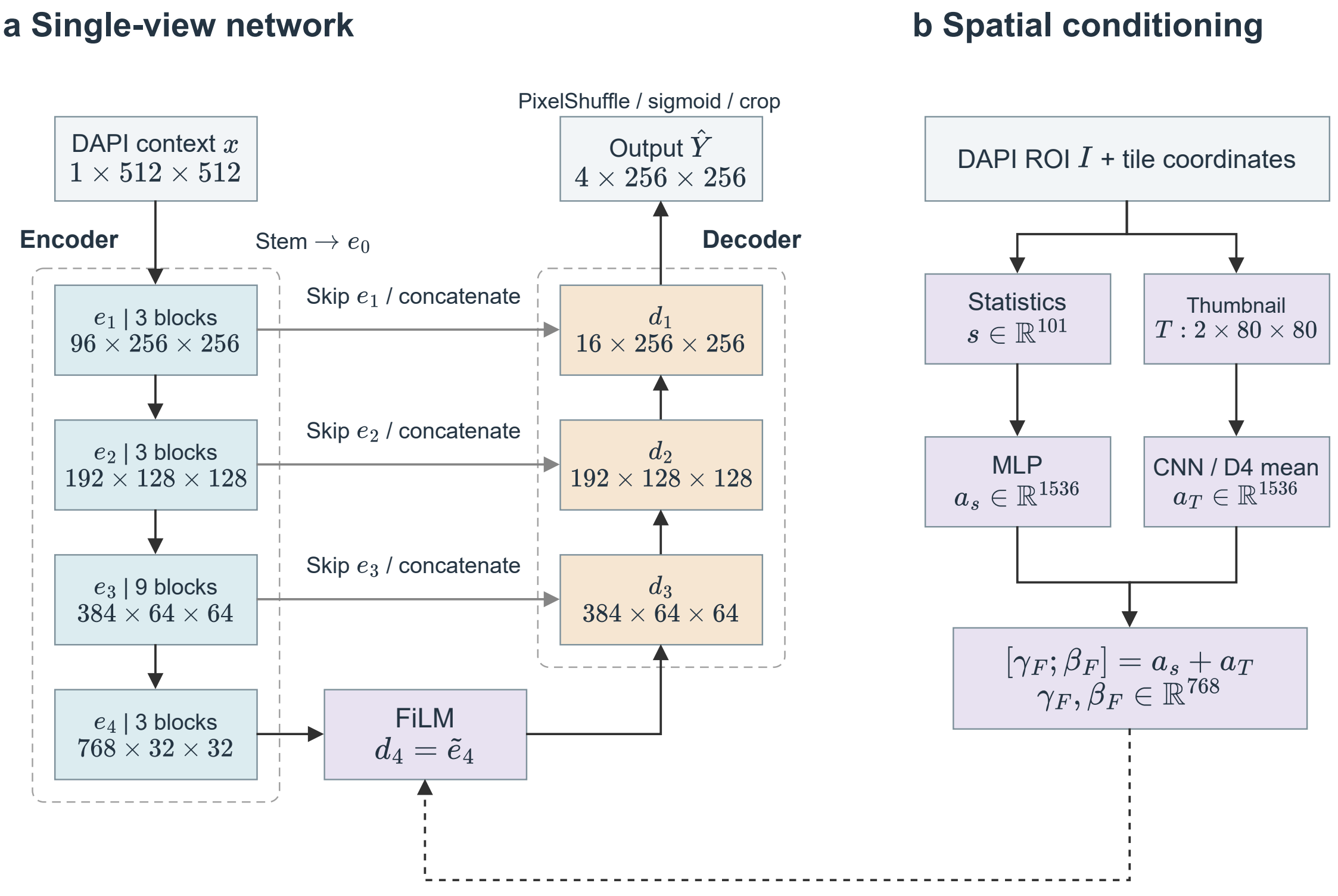

a Single-view network
b Spatial conditioning
PixelShuffle / sigmoid / crop
DAPI context x
$1 \times 512 \times 512$
Output $\hat{Y}$
$4 \times 256 \times 256$
Encoder
Stem $\rightarrow e_0$
Decoder
e_1 | 3 blocks
$96 \times 256 \times 256$
Skip e_1 / concatenate
d_1
$16 \times 256 \times 256$
e_2 | 3 blocks
$192 \times 128 \times 128$
Skip e_2 / concatenate
d_2
$192 \times 128 \times 128$
e_3 | 9 blocks
$384 \times 64 \times 64$
Skip e_3 / concatenate
d_3
$384 \times 64 \times 64$
e_4 | 3 blocks
$768 \times 32 \times 32$
FiLM
$d_4 = \tilde{e}_4$
DAPI ROI I + tile coordinates
Statistics
$s \in \mathbb{R}^{101}$
Thumbnail
$T : 2 \times 80 \times 80$
MLP
$a_s \in \mathbb{R}^{1536}$
CNN / D4 mean
$a_T \in \mathbb{R}^{1536}$
$[\gamma_F; \beta_F] = a_s + a_T$
$\gamma_F, \beta_F \in \mathbb{R}^{768}$